

Mind - Body Interaction

**text in italics will not be displayed to respondents*

Instructions

In this study, we will ask you to review some profiles for developments at a technology company and then answer some questions.

Commented [1]: To reduce deception

You must be at least 18 years old to take part in this study. Your participation is voluntary and you can end the study at any time. There are no known or anticipated risks to you for participating. This study will not benefit you personally. Participation will take approximately 25 minutes. All of your responses will be anonymous.

We will ask questions to ensure you are paying attention, and you cannot go back in this survey, so please take your time before answering.

Consent

I have read the information above and consent to taking part in the study.

Yes, continue to the study.

No, I do not want to take part.

Please enter your Prolific ID below:

Experiment

Mixed between- and within-subjects design:

- 5 (between-subjects, mind: basic control system, agency, emotion, agency > emotion, emotion > agency)
x
- 5 (within-subjects, body: basic, agentic features, emotive features, biomorphic features, anthropomorphic features)

Operationalization: Product development opinions

On the next page, you will read information about some products being developed by a tech company.

Enki Industries is a developer of robotic technologies. Enki creates a variety of robotic bodies to suit various consumer and industrial purposes.

- By definition, **robots** have to be able to **interact with the physical environment**.
- Robots have either **physical bodies** or **physical parts** like levers or wheels.

B-S condition 1: Basic

- Enki's robots are run by **basic control systems**.
- Basic control systems are:
 - **Rules and instructions** such as **software programs** that tell the robot what to do when something specific happens.
 - For example, if the robot encounters the edge of a stair, then a software program is triggered to turn it around to move away from the edge.
- Enki's basic control systems have:
 - **80% navigation** capabilities to **map environments in real time**.
 - **20% observation** capabilities to **keep tabs on their battery status**.
- For example, robots with **basic control systems** can:
 - follow a path.
 - avoid obstacles.
 - track positions.

Commented [2]: Changed to emphasize mind (e.g., observation) instead of body (e.g., movement)

B-S condition 2: Agency

- Enki's robots are run by **artificial intelligence (AI) systems**. AI systems are:
 - **Algorithms** such as **large language models** that allow the robot to converse in human languages or solve complex problems.
 - For example, if the robot encounters a person in the room, then the robot is able to generate language to converse with the person.
- Enki's AI systems have:
 - **80% agency** capabilities to **have their own goals, take independent action to achieve them, and choose their own behavior**.
 - **20% observation** capabilities to **keep tabs on their battery status**.
- For example, robots with **agentic AI systems** can:
 - plan a meeting.
 - organize a calendar.
 - send emails.

Commented [3]: updated

B-S condition 3: Emotion

- Enki's robots are run by **artificial intelligence (AI) systems**. AI systems are:
 - **Algorithms** such as **large language models** that allow the robot to converse in human languages or solve complex problems.
 - For example, if the robot encounters a person in the room, then the robot is able to generate language to converse with the person.
- Enki's AI systems have:
 - **80% emotion** capabilities to **express emotions like empathy and sympathy, detect moods and feelings, and regulate their own feelings**.
 - **20% observation** capabilities to **keep tabs on their battery status**.

Commented [4]: updated

- For example, robots with **emotive AI systems** can:
 - react to situations.
 - express excitement and disappointment.
 - match conversation tone.

B-S condition 4: Agency > emotion

- Enki's robots are run by **artificial intelligence (AI) systems**. AI systems are:
 - **Algorithms** such as **large language models** that allow the robot to converse in human languages or solve complex problems.
 - For example, if the robot encounters a person in the room, then the robot is able to generate language to converse with the person.
- Enki's AI systems have:
 - **80% agency** capabilities to **have their own goals, take independent action to achieve them, and choose their own behavior**.
 - **20% emotion** capabilities to **express emotions like empathy and sympathy, detect moods and feelings, and regulate their own feelings**.
- For example, robots with **80% agentic and 20% emotive AI systems** can:
 - plan a meeting.
 - organize a calendar.
 - send emails.

B-S condition 5: Emotion > agency

- Enki's robots are run by **artificial intelligence (AI) systems**. AI systems are:
 - **Algorithms** such as **large language models** that allow the robot to converse in human languages or solve complex problems.
 - For example, if the robot encounters a person in the room, then the robot is able to generate language to converse with the person.
- Enki's AI systems have:
 - **80% emotion** capabilities to **express emotions like empathy and sympathy, detect moods and feelings, and regulate their own feelings**.
 - **20% agency** capabilities to **have their own goals, take independent action to achieve them, and choose their own behavior**.
- For example, robots with **80% emotive and 20% agentic AI systems** can:
 - react to situations.
 - feel excitement and disappointment.
 - match conversation tone.

Comprehension and attention checks:

1. Which behavior best matches the capabilities of the system? (*mindcomprehension*)
 - a. Matches the emotional tone of a user during an emotional conversation
 - b. Organizes the calendar and sends emails for a user
 - c. Maps the location of trees, dogs, and people in real time

Commented [5]: Emotion and emotion > agency should both select this option.

Commented [6]: Agency and agency > emotion should both select this option

Commented [7]: Basic control should select this option

- d. None of the above
2. Robots have been studied systematically since the middle of last century although the word “robot” as we know it first appeared early last century. If you read this, respond with ‘agree’ for this item. multiple choice: 1 = strongly disagree, 2 = disagree, 3 = somewhat disagree, 4 = no opinion, 5 = somewhat agree, 6 = agree, 7 = strongly agree (*attention 1*)

Next page:

The Enki production team is seeking consumer feedback on several products they have in development. They are interested in your genuine opinions.

For this purpose, you will be asked to review several robot profiles that have **[a/an treatment condition]** on the next few pages.

This means that the robots are run by a system that has [insert specific treatment from list below].

- 80% navigation capabilities to map environments in real time and 20% observation capabilities to keep tabs on their battery status
- 80% agency capabilities to have their own goals, take independent action to achieve them, and choose their own behavior and 20% observation capabilities to keep tabs on their battery status
- 80% emotion capabilities to express emotions like empathy and sympathy, detect moods and feelings, and regulate their own feelings and 20% observation capabilities to keep tabs on their battery status
- 80% agency capabilities to have their own goals, take independent action to achieve them, and choose their own behavior and 20% emotion capabilities to express emotions like empathy and sympathy, detect moods and feelings, and regulate their own feelings
- 80% emotion capabilities to express emotions like empathy and sympathy, detect moods and feelings, and regulate their own feelings and 20% agency capabilities to have their own goals, take independent action to achieve them, and choose their own behavior

Next pages - product specification (within-subjects treatment):

Note. One per page, randomized order. Image for each

Reminder at top of pages: Reminder: Enki’s robot bodies are operated by a/an **[treatment condition]**. They have [mind reminder].

Instructions in between bodies: Next you will review another robot profile.

- *Basic features*

- Proto [randomized 2-digit number]



- *Image:*

- Description and functions:

- **Body**

- Rectangular torso
 - Three wheels to move around
 - Two antennas to **receive wireless signals**

- **Basic sensors**

- **Vision** sensors to see the environment with a camera
 - **Proximity** sensors to know how close or far away objects are

- *Agentic features*

- Proto [randomized 2-digit number]



- *Image:*

- Description and functions:

- **Body**

- Rectangular torso with a **display screen to show its planned directions**
 - Three wheels to move around
 - Magnetic surfaces to **attract and hold objects**

- **Basic sensors**

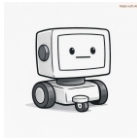
- **Vision** sensors to see the environment with a camera
 - **Proximity** sensors to know how close or far away objects are

- **Unique sensors**

- **Limb-position** sensors to **control where its wheels are** in relation to its torso
 - **Torque** sensors to **know how much force it is using** to move its wheels

- *Emotive features*

- Proto [randomized 2-digit number]



- *Image:*
 - Description and functions:
 - **Body**
 - Rectangular torso with a **display screen to show expressions like pain and pleasure**
 - Three wheels to move around
 - Magnetic surfaces to **attract and hold objects**
 - **Basic sensors**
 - **Vision** sensors to see the environment with a camera
 - **Proximity** sensors to know how close or far away objects are
 - **Unique sensors**
 - **Tactile** sensors to **feel the texture of surfaces** it touches
 - **Nociceptor** sensors to **experience pain when touching** a hot or cold surface
 - *Biomorphic features*
 - Proto [randomized 2-digit number]
-
-
- *Image:*
 - Description and functions:
 - **Body**
 - Symmetrical **bioform** with a **head, neck, and face to show expressions**
 - Multiple limbs to **move around horizontally**
 - Limbs with **gripping tips to grasp and hold objects**
 - **Basic sensors**
 - **Vision** sensors to see the environment with a camera
 - **Proximity** sensors to know how close or far away objects are
 - **Unique sensors**
 - **Proprioceptive** sensors to **know where its limbs are** in relation to its torso
 - **Scalar** sensors to **know how much pressure is on its limbs** when it holds objects
 - *Anthropomorphic features*
 - Proto [randomized 2-digit number]



- *Image:*
- Description and functions:
 - **Body**
 - Symmetrical **humanoid** body with head, neck, face, and **forward-facing eyes to show expressions and make eye contact**
 - Two legs to **move around upright**
 - Two arms with **dexterous hands to grasp and hold objects**
 - **Basic sensors**
 - **Vision** sensors to see the environment with a camera
 - **Proximity** sensors to know how close or far away objects are
 - **Unique sensors**
 - **Limb-position** sensors to **control where its arms and legs are** in relation to its torso
 - **Tactile** sensors to **feel the texture of surfaces** it touches

Entity Perceptions (Within-Subjects)

Note. Items are presented for each entity (5 evaluations per participant). Randomized order of scales, except comprehension always presented first. Items within each scale are randomized. Each scale is on its own page.

Reminder at top of instruction pages, except comprehension check: Reminder: Enki's robot bodies are operated by the [treatment condition]. They have [mind reminder]. This body has [unique sensors] and [unique body feature]. [image]

Comprehension checks

1. Which of the following physical features does [target] have? (*featurecheck*)
 - a. Dexterous hands
 - b. Gripping tips
 - c. Torque (force) sensors
 - d. Nociceptor (pain) sensors
 - e. Antennas
2. How unclear or clear is your sense of [target] with [mind]? (sliding scale: -3 = very unclear, 3 = very clear) (*comboease*)
3. How much do you dislike or like [target] with [mind]? (sliding scale: -3 = very much dislike, 3 = very much like) (*likecombo*)
4. To what extent do you feel that [target] with [mind] is creepy or inviting? (sliding scale: -3 = very creepy, 3 = very inviting) (*uncanny*)

Commented [8]: Should be selected for the anthropomorphic robot

Commented [9]: Should be selected for the bioform robot

Commented [10]: Should be selected for the agency robot

Commented [11]: Should be selected for the emotion robot

Commented [12]: Should be selected for the basic robot

Commented [13]: New to be used as potential covariates

Mind perception (MPC)

To what extent does [target] have the capacity for each of the following?

sliding scale: 0 = not at all, 100 = very much

1. experiencing emotions (*MP1*)
2. having feelings (*MP2*)
3. thinking analytically (*MP3*)
4. being rational (*MP4*)
5. having their own motivations (*agency1*)
6. controlling themselves (*agency2*)

Moral treatment (AITC)

To what extent do you agree or disagree with the following statements?

sliding scale: 1 = strongly disagree, 7 = strongly agree

1. [Target] deserves to be treated with respect. (*MCE1*)
2. [Target] deserves to be included in the moral circle. (*MCE2*)
3. Physically damaging [target] without their consent is wrong. (*MCE3*)
4. Re-programming [target] without their consent is wrong. (*MCE4*)
5. Torturing [target] is wrong. (*MCE5*)
6. The welfare of robots like [target] is one of the most important social issues in the world today. (*MCE6*)

Specific positive (patency) and negative (agency) rights

As part of a change in **how society regulates AI**, experts in moral philosophy, AI, and law are debating **legal frameworks** that would entitle [target] to **certain legal rights**. Below, we list the **rights that [target] could be given**.

To what extent would you agree or disagree with [target] being given the following rights?

sliding scale: 1 = strongly disagree, 7 = strongly agree

1. [Target] has the right to vote in elections. (*vote_neg*)
2. [Target] has the right to own property. (*property_neg*)
3. [Target] is allowed to engage in financial transactions. (*money_neg*)
4. [Target] is allowed to enter into legal contracts. (*contracts_neg*)
5. [Target] has the right to choose their own job. (*job_neg*)
6. [Target] has legal status to act like how corporations can act as legal persons. (*legal_neg*)
7. [Target] has legal status to be protected like how humans are protected as natural persons. (*legal_pos*)
8. [Target] is allowed to copyright the work they produce. (*copyright_pos*)
9. [Target] is paid by their employer for the work they do. (*remuneration_pos*)
10. [Target] has the right to give consent before they are re-programmed. (*consent_pos*)

Commented [14]: updated

11. [Target] has the right to give consent before they are shut down. (*shutdown_pos*)
12. [Target] has the right to be protected from cruel treatment by developers and users. (*cruelty_pos*)

Reciprocity (*inspired by Perugini et al., 2003*)

To what extent could [target] behave in the following ways?

sliding scale: 0 = not at all, 100 = very much

1. [target] goes out of their way to help somebody who has been kind to them before. (*posrec1*)
2. If someone asks [target] politely for information, [target] is really happy to help them. (*posrec2*)
3. If someone does a favor for [target], [target] is ready to return it. (*posrec3*)
4. The way [target] treats others depends on how they treat [target]. (*negrec1*)
5. If somebody is impolite to [target], [target] becomes impolite. (*negrec2*)
6. If somebody offends [target], [target] will offend them back. (*negrec3*)

Intrinsic worth/personhood (*adapted from Johnson et al., 2015, used in DC paper*)

To what extent would you endorse the following statements about [target]?

sliding scale: 0 = not at all, 100 = very much

1. [target] is a person. (*person*)
2. [target] has a soul. (*soul*)
3. [target] has an inner essence. (*essence*)
4. [target] is worthy of moral care. (*moralworth*)

Checks

Note. Randomized order

We have a few more questions for you.

1. Which tech company did you review products for? (*companycheck*)
 - a. Enki
 - b. Anthropic
 - c. Hitachi
 - d. Talos
 - e. Yanta
 - f. Luban
2. True or False: Each of the robots that you read about had [a basic control system, an AI system]. (*AIcheck*)
 - a. True
 - b. False
3. How many robots did you read about? (*robotnumbercheck*)

- a. 0
- b. 1
- c. 2
- d. 3
- e. 4
- f. 5
- g. 6

Additional Questions

Note. Randomized items

Beliefs (adapted from mind perception targets and declarative belief statements & measurement scale from climate change and health beliefs updating literature - [Vlasceanu](#))

How accurate or inaccurate do you think each of the following statements is?

sliding scale: 0 = not at all accurate, 100 = extremely accurate

1. Machines cannot have feelings. (BEM1)
2. Als can experience emotions. (BEM2; R)
3. Als will never be able to feel sad. (BEM3)
4. Someday, robots could feel happy. (BEM4; R)
5. It is impossible for any machine, even AI, to ever have true feelings. (BEM5)
6. Humans are the only entities that can experience emotions. (BHM1)
7. Humans are different from all other animals. (BHM2)
8. Humans are the only entities that can be creative. (BHM3)
9. Free will is a basic part of human nature. (BHM4)
10. Other animals can choose whether or not to act on their own goals. (BNHM1)
11. Als can make their own decisions about what they want to do. (BNHM2)
12. (foil) Human activities are causing climate change. (Bfoil1, note = accurate)
13. (foil) Climate change poses a serious threat to humanity. (Bfoil2, note = accurate)
14. (foil) Corporal punishment is associated with lower Intelligence Quotient in children. (Bfoil3, note = accurate)
15. (foil) Children can outgrow peanut allergies. (Bfoil4, note = accurate)
16. (foil) Fish can feel pain. (Bfoil5, note = accurate)
17. (foil) Farmed animals have better lives on small farms than factory farms. (Bfoil6, note = accurate-ish)

AI Experiences and Demographics

Next, you will see questions about your own experience with robots/Als.

Personal experience

10

- Commented [15]: machines can't feel
- Commented [16]: Established factor loading? If thes ...
- Commented [17]: No established loading! I like the ...
- Commented [18]: _Marked as resolved_
- Commented [19]: _Re-opened_
- Commented [20]: machines can't feel
- Commented [21]: Established factor loading? If thes ...
- Commented [22]: No established loading! I like the ...
- Commented [23]: _Marked as resolved_
- Commented [24]: _Re-opened_
- Commented [25]: humans are special
- Commented [26]: machines can't feel
- Commented [27]: Established factor loading? If thes ...
- Commented [28]: No established loading! I like the ...
- Commented [29]: _Marked as resolved_
- Commented [30]: _Re-opened_
- Commented [31]: humans are special
- Commented [32]: machines can't feel
- Commented [33]: Established factor loading? If thes ...
- Commented [34]: No established loading! I like the ...
- Commented [35]: _Marked as resolved_
- Commented [36]: _Re-opened_
- Commented [37]: humans are special
- Commented [38]: machines can't feel
- Commented [39]: Established factor loading? If thes ...
- Commented [40]: No established loading! I like the ...
- Commented [41]: _Marked as resolved_
- Commented [42]: _Re-opened_
- Commented [43]: Item from Rakos et al., 2008
- Commented [44]: free will
- Commented [45]: machines can't feel
- Commented [46]: Established factor loading? If thes ...
- Commented [47]: No established loading! I like the ...
- Commented [48]: _Marked as resolved_
- Commented [49]: _Re-opened_
- Commented [50]: free will
- Commented [51]: machines can't feel
- Commented [52]: Established factor loading? If thes ...
- Commented [53]: No established loading! I like the ...
- Commented [54]: _Marked as resolved_
- Commented [55]: _Re-opened_
- Commented [56]: free will
- Commented [57]: machines can't feel
- Commented [58]: Established factor loading? If thes ...
- Commented [59]: No established loading! I like the ...
- Commented [60]: _Marked as resolved_

1. I am interested in buying more smart devices with some ability to detect their environment and network with other devices. If you read this, respond with 'Somewhat Agree' for this item. (*attention2*)

2. **Als at home** (*own*)

Do you own AI or robotic devices that can detect their environment and respond appropriately? For example, you might own...

- a home assistant that can detect whatever you say and respond in kind
- a home assistant that can control the temperature and ambient lighting based on feedback it receives from the environment
- a robotic vacuum cleaner that decides where to move by itself
- a self-driving car that monitors its environment and drives itself
- a companion robot that can interact with you socially

No

Yes

3. **Als at work** (*work*)

Do you work with AI or robotic devices at your job? For example, you might work with a robot/AI...

- for research or scientific purposes
- for information or fact checking
- for facial recognition
- for healthcare diagnoses
- for making social work or legal decisions
- for aerial photography or mapping
- for disaster response
- for industrial production
- for creative arts

No

Yes

4. **AI companions** (*companion*)

Do you have any AI friends or companions that you interact with regularly? For example, you might talk to a chatbot about your thoughts or feelings.

No

Yes

Not sure

5. **Smart devices** (*smart*)

Do you own a smart device that has some ability to detect its environment and network with other devices but that cannot respond to everything you might say or that requires you to pre-program its routines? For instance, you might own...

- a smartphone
- a smart TV
- smart speakers
- a smart coffee maker

- a smart toaster
- smart light bulbs

No
Yes

6. How often do you interact with AI or robotic devices that respond to you and that can choose their own behavior? (*fint*)

- 0 - Never or Not applicable
- 1 - Only on rare occasions
- 2 - A few times a year
- 3 - Monthly
- 4 - Weekly
- 5 - Daily

7. How often do you read or watch robot/AI-related stories, movies, TV shows, comics, news, product descriptions, conference papers, journal papers, blogs, or other material? (*fexp*)

- 0 - Never or Not applicable
- 1 - Only on rare occasions
- 2 - A few times a year
- 3 - Monthly
- 4 - Weekly
- 5 - Daily

Finally, you will see some questions about your personal characteristics.

Age

What is your age? (*integer slider 18-100*)

Gender

What is your gender?

Man
Nonbinary
Woman
Prefer not to say

Race

Which of the following best describes you?

American Indian, Native American or Alaskan Native
Asian, Asian-American or Pacific Islander
Black or African-American
Dual or Multiple Ethnicities
Hispanic or Latine-American

White or European-American (non-Hispanic)
 Other
 Prefer not to say

Family income

What is your family's net household income per year?

\$0-20,999
 \$21,000-40,999
 \$41,000-60,999
 \$61,000-80,999
 \$81,000-99,999
 \$100,000-149,999
 \$150,000-199,999
 \$200,000-249,999
 \$250,000+
 Prefer not to say

Own education level

What is the highest degree that you have completed?

None or not applicable
 High School Diploma or GED
 Associate's Degree
 Bachelor's Degree
 Master's Degree
 Doctoral Degree (for example law, medicine, philosophy, arts, or science)
 Prefer not to say

Political orientation

How would you describe your political views? (1 = very liberal, 3 = moderate, 5 = very conservative)

Religion

What is your present religion, if any?

Buddhist
 Christian-Catholic (for example Roman Catholic and Orthodox)
 Christian-Protestant (for example Episcopalian, Presbyterian, United Church of Christ, Methodist, Baptist, Lutheran)
 Hindu
 Jain
 Jewish
 Muslim
 Shinto
 Sikh
 Atheist (do not believe in God)

Agnostic (not sure if there is a God)
 No religion
 Something else (if you answered 'something else' for religion, please specify).
 Prefer not to say

Diet

Which of these best describes what you currently eat?

Meat-eater (no restrictions on animal products)
 Pescatarian (no chicken, beef, or any meats other than fish)
 Vegetarian (no meat)
 Vegan (no animal products)
 Other restrictions on animal products

Debriefing

Thank you for taking part in this study!

The purpose of this study is to understand how people treat AIs and robots that are made up of different mind and body systems. To do this, we had some people read about robots that are controlled by a basic control system like a robot vacuum and some people read about robots controlled by an AI system like a large language model. These AI systems were defined either by mostly agentic capabilities like having goals or mostly emotive capabilities like expressing emotions. Some people read about systems that were partly agentic and partly emotive. Everyone then answered the same questions about all the same fictitious robots being developed at a fictitious company. The company and robot profiles were created for the purpose of this study but the robot and AI capabilities were inspired by real-life technologies. We hope that this survey will help us understand how people treat and think about the rights of robots and AIs with varying capabilities.

This study was conducted by Sentience Institute, a think tank that researches social and technological change. You can find out more about our work [here]<https://sentienceinstitute.org>. If you are interested in learning the results of the study once it is completed, or if you have any other concerns or queries, please feel free to contact us at info@sentienceinstitute.org.

If you would like to leave any feedback about the study, please feel free to do so in the box below.

Please leave any additional feedback you have here. (*open-ended; feedback*)

When you are ready, please click the 'Complete survey' button below to ensure your responses are submitted. You will be redirected automatically back to Prolific with a completion code.